ZKSF
Zero Kelvin Simulation Foundry

ZHF-v0.1
2026-10-28T11:40:55+00:00

Certificate of Hardware Fidelity

Certificate ID
c752e4db92bb418a

Measured hardware fidelity: 0.4160 vs. exact ideal

ENGINE
qpu.aqt.ibex
SHOTS
100
HELLINGER FIDELITY
0.4160

METHOD
quantum hardware (AQT IBEX Q1 (trapped ion))
DEVICE
Ibex-Q1
VERIFICATION MODE
direct

MEASURED OUTCOMES (top)

000	32 shots
100	13 shots
001	13 shots
111	12 shots
110	10 shots
010	8 shots

HOW THIS CERTIFICATE WAS PRODUCED

• This certificate reports ZHF-v0.1, the hardware fidelity protocol: how closely a real quantum processor's measured output matches the ideal distribution for the same circuit, as opposed to the accuracy of a classical approximation.

• Verification mode is direct: the same circuit was additionally simulated exactly on a classical computer (feasible here because it uses at most 24 qubits). The reported Hellinger fidelity is the statistical overlap between the hardware's measured counts and that exact ideal distribution, on a scale from 0 (no overlap) to 1 (identical).

• Counts are raw hardware output: no error mitigation, post-selection, or readout correction has been applied. Any deviation from the ideal distribution reflects the physical noise of the device.

Independently verifiable at:
https://api.zksf.org/certify/c752e4db92bb418a

Circuit SHA-256 99b41c00d8345c3a2075bce453e75bcb43a1a6af0dc51855bf759b4482643fee